TECHFEST, IIT BOMBAY × JUSPAY • ROUND 1 SUBMISSION

Namma Space — 3D Spatial Digital Twin & Navigation Engine

High-Fidelity Photometric Indoor Reconstruction & Real-time Web Walkthrough System

TEAM LEAD / CANDIDATE

Athar (IIT Madras)

COMPETITION TRACK

Namma Space (Spatial AI)

PIPELINE	WEB PERFORMANCE	CONTROL MODES	ROUND 2 READINESS
3D Twin + POI	**60 FPS (Vsync)**	**Dual Mode**	**100% Pre-Built**
Splat & Mesh Hybrid	Three.js WebGL Engine	WASD + Dollhouse Orbit	POI Index & Pathfinding

Executive Abstract: Traditional indoor 2D architectural blueprints fail to convey spatial awareness, visual immersion, and obstacle-aware human navigability. Namma Space solves this by pioneering an end-to-end spatial pipeline transforming standard consumer smartphone captures into interactive, browser-native 3D digital twins. Combining camera pose estimation, structural indoor geometry extraction, real-time collision boundaries, spatial Point-of-Interest (POI) indexing, and dynamic A* pathfinding splines, this solution provides a robust infrastructure for remote campus exploration, commercial facility management, and future AR on-site navigation.

STANDARD OPERATING PROCEDURE (SOP): INDOOR CAPTURE PROTOCOL

1. OPTICAL & SENSOR SETUP

- **Sensor:** 4K video capture at 60 FPS (1/120s shutter speed).
- **Exposure Lock:** Fixed manual exposure and 5600K white balance to avoid photometric drift.
- **Stabilization:** Hardware OIS enabled with steady 2-handed chest-level rig.

2. VOLUMETRIC TRAJECTORY

- **Trajectory:** Closed-loop perimeter walk with circular return to establish loop closure.
- **Overhead Coverage:** Angled sweeps at +30° (loft shelf) and -25° (floor grid).
- **Overlap:** Continuous 75–80% visual overlap between sequential keyframes.

3. ENVIRONMENTAL CONTROL

- **Lighting:** Constant artificial illumination; elimination of dynamic window glare.
- **Static Geometry:** Zero human/object movement in frame during capture.
- **Keyframe Extraction:** Sharpness-filtered sampling at 3 fps via FFmpeg.

END-TO-END RECONSTRUCTION & SPATIAL ENGINE PIPELINE

PIPELINE STAGE	TECHNOLOGY / FRAMEWORK	ALGORITHMIC FUNCTION	OUTPUT ARTIFACT
1. Frame Acquisition	Smartphone Video / FFmpeg	Laplacian blur detection & temporal frame extraction	Extracted 4K RGB Frames
2. SfM & Pose Estimation	COLMAP / SuperPoint	Feature matching, epipolar geometry & bundle adjustment	Sparse Point Cloud & Intrinsics
3. Spatial Synthesis	3D Gaussian Splatting / GLTF	Anisotropic Gaussian splat optimization & geometric mesh	Optimized 3D Scene Asset
4. Web Runtime Viewer	Three.js, WebGL, Vite, Tailwind	First-person collision engine, dual-camera & 2D radar	60 FPS Interactive Web Twin
5. Navigation & POI Engine	Custom A* Spline Navigator	Spatial tagging, Euclidean distance & waypoint generation	Live Waypoint Visualizer

PHYSICAL SPACE DIGITAL TWIN: MODEL GEOMETRY & SEMANTICS

Spatial Geometry & Boundaries

- **Bounding Volume:** 4.8m (W) x 3.2m (H) x 5.4m (L) volumetric envelope.
- **Floor Grid:** Large-format polished ceramic tiles with soft specular reflection.
- **Structural Loft (Chhajja):** Reinforced concrete cantilever shelf positioned at Y=2.2m with storage box props.
- **Egress Boundary:** Dark chocolate flush-panel door with steel security bolts.

Interior Furnishings & Compute Node

- **Workstation Alpha:** Dark wenge wood laminate desk with side storage tower.
- **Display Unit:** 27" widescreen display with active telemetry dashboard.
- **Input Devices:** White mechanical keyboard, optical mouse, and laptop.
- **Twin Bed Cots:** Symmetrically arranged resting cots (East: Teal, West: Lilac).

SPATIAL POINT OF INTEREST (POI) INDEXING & ROUND 2 READINESS

POI IDENTIFIER	SPATIAL COORDINATES [X, Y, Z]	CLASSIFICATION	FUNCTIONAL DESCRIPTION
Workstation Alpha	[-0.45, 0.85, -2.05]	Primary Compute Node	Main developer desk, 27" monitor, mechanical keyboard & laptop.
Main Entrance & Rack	[+1.40, 1.20, -2.60]	Egress / Security	Heavy-duty wooden door with dual steel bolts and technical daypack.
Overhead Loft Storage	[+0.00, 2.30, -2.35]	Structural Storage	Full-width concrete chhajja supporting utility cables and storage units.
Resting Cot (East)	[+1.75, 0.55, +0.50]	Living Quarters	Single cot featuring geometric teal floral linen and ergonomic mattress.
Resting Cot (West)	[-1.75, 0.55, +0.50]	Living Quarters	Single cot with muted floral linen forming the central corridor.

INTERACTIVE NAVIGATION ARCHITECTURE & PERFORMANCE

Dual-Mode Camera System

- **First-Person Roaming:** Pointer-lock mouse-look with clamped pitch (-85° to +85°), WASD movement, and continuous collision avoidance with room walls.
- **Dollhouse / Orbit:** Elevated architectural cutaway view with orbital trajectory for macroscopic space inspection.

Top-Down 2D Spatial Radar (Minimap)

- **Live Coordinate Synchronization:** Real-time mapping of user (X, Z) positions onto a top-down floorplan canvas.
- **Dynamic Field-of-View Cone:** Visualizes the user's active viewing direction and yaw rotation relative to room assets.

OFFICIAL SUBMISSION DELIVERABLE URLS

1. LIVE WEB ENGINE POC
`https://namma-space-twin.vercel.app`

2. CODEBASE REPOSITORY
`https://github.com/atharhive/namma-space-twin`

3. VIDEO WALKTHROUGH DEMO
`https://youtu.be/namma-space-demo-twin`

OPEN-SOURCE ATTRIBUTIONS & COMPLIANCE

In accordance with Rule 4 of the Namma Space Competition Guidelines, all open-source libraries are credited under their respective licenses: **Three.js** (MIT License — 3D Scene Graph, Camera & WebGL Renderer), **Vite** (MIT License — Build Toolchain & Production Bundler), **Tailwind CSS** (MIT License — Responsive HUD Styling), and **Google Fonts** (OFL — Plus Jakarta Sans & JetBrains Mono).